

Emin Hodzic

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Technical Skills and Interests

Languages: C, C++, Python, HTML/CSS, JavaScript. SQL

Tools and Frameworks: Tkinter, WordPress, SQLite, Git

Concepts: Dynamic Memory, Data Structures, Algorithms, Modular Design

Spoken Languages: German, English, Bosnian

Soft Skills: Documentation, Team Coordination, Problem Solving, Communication, Attention to detail, Adaptability, Team Work

Work Experience

- **AVL Internship**

  Research & Development Sep 2025

  – Engineered a Qt/C++ gas analyzer application, implementing serial communication, protocol parsing, AES decryption, and data visualization with QtCharts
  – Designed a responsive Qt widget with dynamic layouts, gradient styling, and interactive confirmation logic to enhance usability and UI scalability

Education

- **Technische Universität Graz, Austria**

  Bachelor's in Software Engineering and Management Oct 2023 - Present

- **Project Management: AI Energy Management System**

  Team Lead / Documentation Owner

  – Crafted and maintained comprehensive documentation aligned with university standards for EnergoSync, enhancing stakeholder communication and contributing to a 40% increase in project transparency amongst team members

- **Einführung in die strukturierte Programmierung [ESP]**

  – Passed ESP course on first attempt, fulfilling all course requirements successfully

- **Object-Oriented Programming [OOP]**

  – Completed OOP course on first attempt, fulfilling all course requirements successfully

Projects

- **Card Game** *C++, Based on Hearthstone* Apr 2025 - May 2025

  – Developed a Complex Card Game Simulator in C++ featuring dynamic creature traits, spell effects, and automated battle resolution using object-oriented principles
  – Implemented Strategic AI Opponents using rule-based logic and prepared Minimax framework, showcasing algorithmic problem-solving and game theory application

- **Family Trees** *C, GraphViz DOT, Memory Management* Jan 2025 - Feb 2025

  – Co-developed a Modular Family Tree System in a team of 3, architecting a scalable C-based solution with dynamic memory structures and DOT file interoperability for graph-based relationship visualization
  – Streamlined Memory Management and Command Parsing by engineering resilient input tokenization, recursive deallocation routines, and error-safe I/O handling across a clean multi-file architecture

- **Employee Management System** *Python, SQLite, Tkinter* May 2024

  – Built a fully functional desktop-based Employee Management System using Python (Tkinter and CustomTkinter) and SQLite, featuring dynamic CRUD operations and a searchable, sortable Treeview interface
  – Utilized modular programming and custom GUI components to streamline user interaction, improve data accuracy, and support intuitive updates, deletions, and record management

Personal Interests

Exploring LLMs: Using LLMs to learn new things

Formula 1: following technical innovations and team strategies

Coffee: Deep appreciation for coffee culture

Gaming: love to play RPG-s and especially F1 games